JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Basic Science & Humanities Department
2023-24 (Semester)

Education to Eternity

MID SEMESTER EXAMINATION

Course: B. Tech. **Subject:** Basic Electrical & Electronics Engg.
Subject Code: CS1T003/AI1T003/DS1T003 **Year/Semester:** I Semester (First Year)

Max Marks: 20 **Date: - 31/10/2023** **Duration:- 1 Hr**

Instructions to the Students:
1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1.Define fundamentals of electrical system and choose measuring instruments for measurement of electrical quantities & describe the concept of PN junction diode and its characteristics.
CO2. Classify wiring system and compare energy resources for electrical energy generation & elaborate the transistor configuration in CE, CB & CC mode.
CO3. Plan and organize the utilization of energy resources of electrical system & apply transistor characteristics to construct amplifiers devices.
CO4. Compare different sources of electrical system & distinguish various logic gates and simplify the Boolean's equations.
CO5. Justify the utilization of various electrical and electronics components into electrical and electronics circuitries.
CO6. Construct various circuits using resistors, capacitors, inductors, PN junction diode, Zener diode, transformers, transistors and logic gates.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/ CO	Marks
Q.1	Define one coulomb charge.	1/CO1	1
Q.2	What is Electrical wiring.	2/CO1	1
Q.3	Why is inductor called passive element?	1/CO2	1
Q.4	Write the equation for Lenz's law.	2/CO2	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	List the factors on which resistance depends?	1/CO1	2
Q.6	How much work will be done by an electric energy source with a potential difference of 3kV that delivers a current of 1A for 1 minute?	1/CO1	2
Q.7	What are different advantages of using cleat type of wiring system?	2/CO2	2
Q.8	Mention different applications of variable inductors?	2/CO3	2

SECTION C

1. **Solve any two questions.**

Q.9	The temperature coefficient of resistance of a wire is 12.5 x 10^{-4} per degree Celsius at 300k. The resistance of the wire is 1Ω. The temperature at which the resistance is 2Ω is?	3/CO5	4
Q.10	Explain in detail the types of variable capacitors.	3/CO2	4
Q.11	Distinguish between Self and Mutual Inductance.	4/CO4	4